LAB 3

SIMULATION OF NETWORK DEVICES USING CISCO PACKET TRACER

Objectives:

- To simulate different network devices using Cisco Packet Tracer.
- To understand the function and working of basic networking devices.

Software Required:

- Cisco Packet Tracer

Theory:

Hub

A hub is a basic networking device used to connect multiple computers in a network. It works at the Physical Layer of the OSI model. When a hub receives data from one device, it broadcasts the data to all other connected devices without checking the destination. Because of this, unnecessary traffic is created in the network and performance is reduced. Hubs are simple and inexpensive but are rarely used in modern networks.

- ✓ Works at Data Link Layer (Layer 2)
- ✓ Uses MAC address table
- ✓ Reduces network traffic

Switch

A switch is an intelligent networking device used in local area networks. It works at the Data Link Layer of the OSI model and uses MAC addresses to forward data. Unlike a hub, a switch sends data only to the destination device, which reduces network traffic and improves speed. Switches are widely used in modern networks due to their efficiency and better performance.

- ✓ Works at Layer 2
- ✓ Filters traffic using MAC address
- ✓ Connects two LAN segments

Bridge

A bridge is used to divide a large network into smaller segments. It works at the Data Link Layer and filters data based on MAC addresses. By separating network segments, a bridge reduces traffic and improves overall network performance. It allows communication between different LAN segments while controlling unnecessary data flow.

- Works at **Layer 2**
- Filters traffic using MAC address
- Connects two LAN segments

Router

A router is a networking device that connects two or more different networks. It works at the Network Layer of the OSI model and routes data using IP addresses. Routers determine the best path for data using routing tables. They are commonly used to connect local networks to the internet and provide secure and efficient data transmission.

- Works at Network Layer (Layer 3)
- Uses routing table
- Connects LAN to WAN

Repeater

A repeater is a device used to extend the distance of a network. It works at the Physical Layer of the OSI model. When signals become weak over long distances, a repeater regenerates and strengthens them before forwarding. Repeaters help in maintaining signal quality but do not filter or manage data traffic.

- Works at Physical Layer (Layer 1)
- Boosts signal strength

Process:

A. **Simulation Steps for hub**

i. Drag a Hub from Network Devices
ii. Connect multiple PCs using Copper Straight-Through cable
iii. Send data between PCs

B. Simulation Steps for switch

i. Place a Switch
ii. Connect PCs to the switch
iii. Send data between PCs

C. Simulation Steps for bridge

i. Place two hubs or switches
ii. Connect them using a Bridge
iii. Send data between segments

D. Simulation Steps for router

i. Place a Router
ii. Connect two different networks
iii. Assign IP addresses

iv. Send data across networks

E. Simulation Steps for repeater

i. Place a Repeater between two devices
ii. Connect cables
iii. Send data

Output

HUB

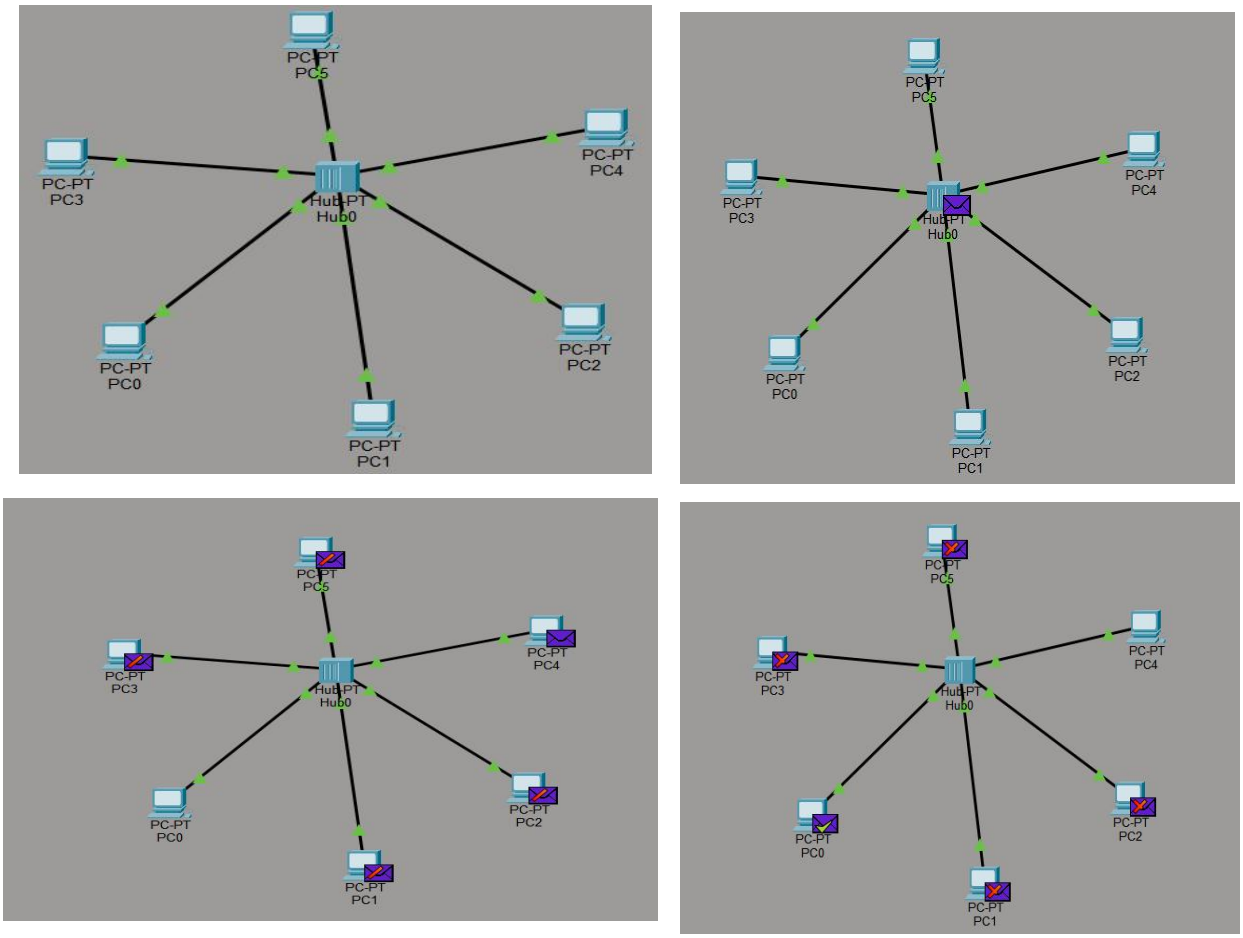


Fig:Hub collecting the message from the source and broadcasting it to all connected computers, where only the intended destination computer accepts the message.

Configuration Table

Device Name	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.1	255.255.255.0
PC1	FastEthernet0	192.168.1.2	255.255.255.0
PC2	FastEthernet0	192.168.1.3	255.255.255.0

PC3	FastEthernet0	192.168.1.4	255.255.255.0
PC4	FastEthernet0	192.168.1.5	255.255.255.0
PC5	FastEthernet0	192.168.1.6	255.255.255.0

In this simulation, a Star topology was created using a Hub. PC0 is connected to the Hub- PT, likewise PC1, PC2, PC3, PC4, and PC5 are connected to the remaining ports of the Hub using Copper Straight-Through cables. When a PDU was sent from PC5 to PC0, the Hub received the packet and broadcasted it to all connected devices (PC4, PC3, PC2, PC1, and PC0). The intended recipient accepted it, while others rejected it.

SWITCH

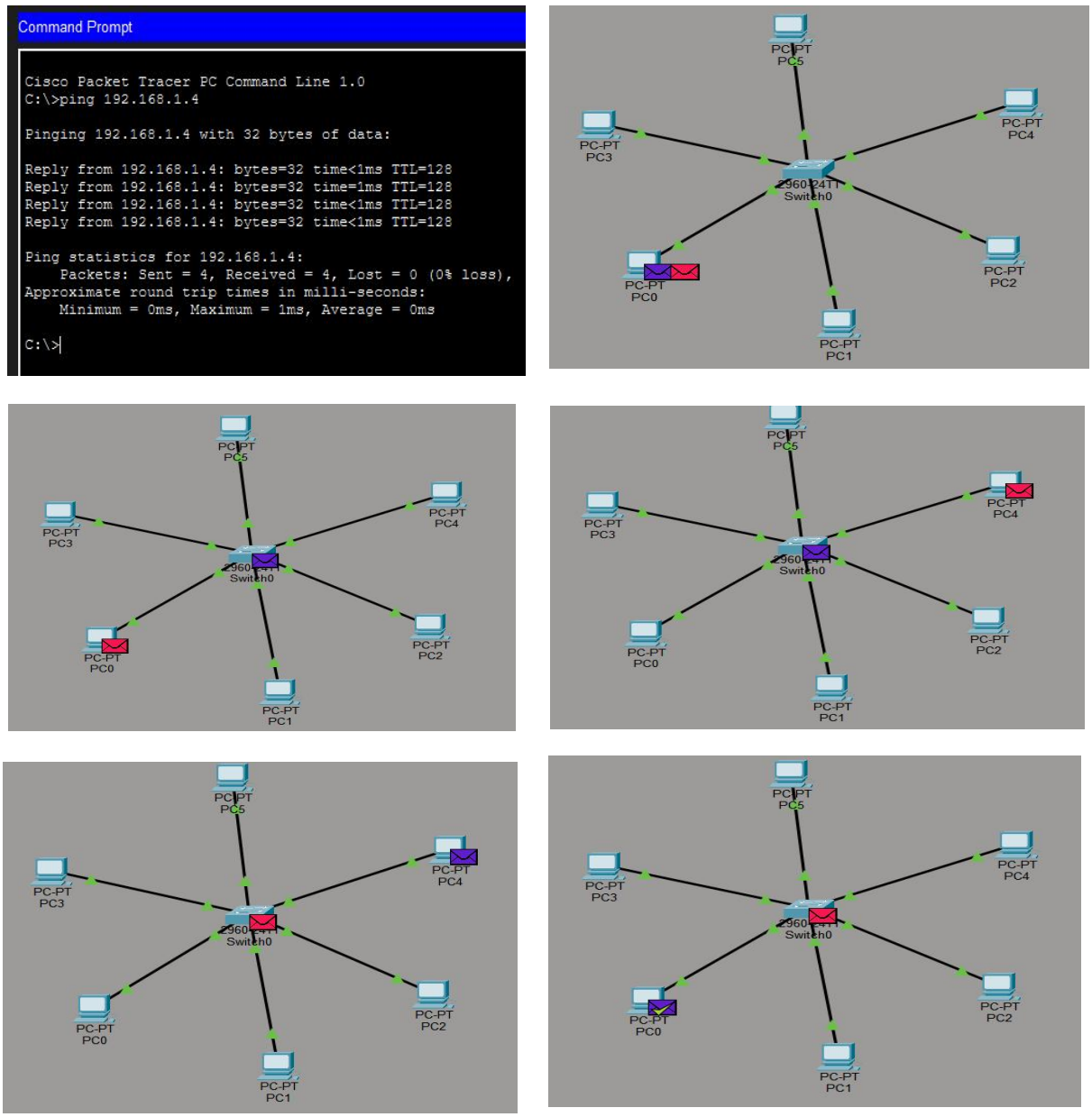


Fig: Switch receiving data from the source and forwarding it only to the intended destination computer using MAC address mapping.

Configuration Table

Device Name	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.10	255.255.255.0
PC1	FastEthernet0	192.168.1.11	255.255.255.0
PC2	FastEthernet0	192.168.1.12	255.255.255.0
PC3	FastEthernet0	192.168.1.13	255.255.255.0
PC4	FastEthernet0	192.168.1.14	255.255.255.0
PC5	FastEthernet0	192.168.1.15	255.255.255.0

A Switch was used to connect multiple devices. PC0, PC1, and PC2 are connected to the Switch 2960-24TT, likewise PC3, PC4, and PC5 are connected to the switch ports. Unlike the Hub, once the Switch learned the MAC addresses, the message flow was observed to be unicast. The packet was sent directly from the source to the destination without disturbing other ports.

BRIDGE

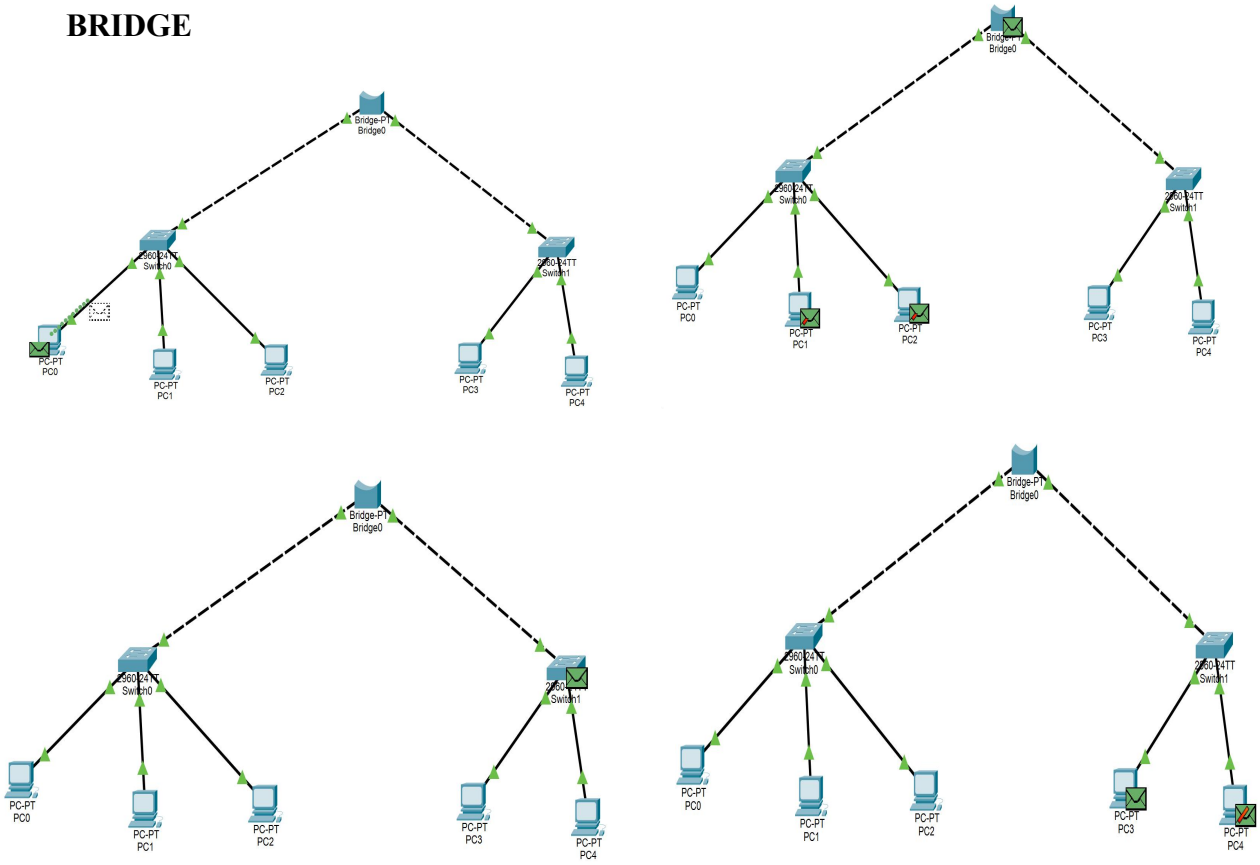
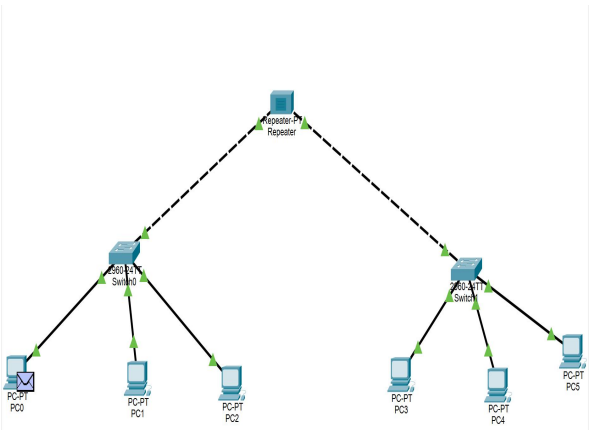


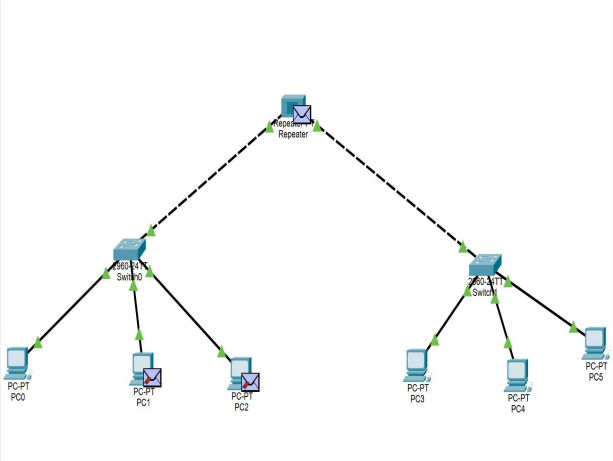
Fig: Bridge receiving data from the source and forwarding it only to the intended destination network segment by using MAC address filtering.

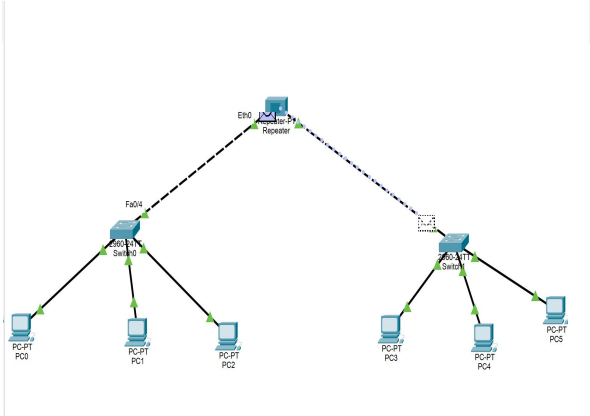
Device Name	IP Address	Subnet Mask
Switch0	192.168.1.2	255.255.255.0
Switch1	10.10.10.2	255.255.255.0
Bridge0	192.168.1.1	255.255.255.0

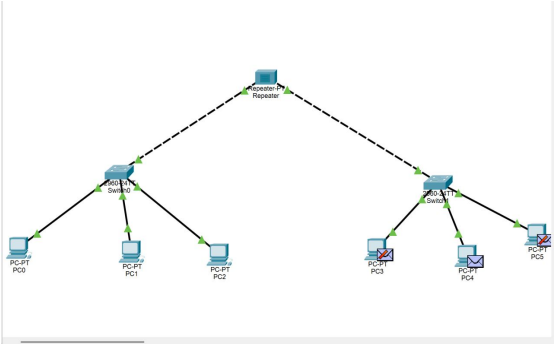
Two network segments were connected using a Bridge. The left segment switch is connected to Bridge-PT Port 0, likewise the right segment switch is connected to Bridge- PT Port 1. The simulation demonstrated how the Bridge filters traffic. It only allows traffic to cross to the other segment if the destination address belongs to a device on that side.

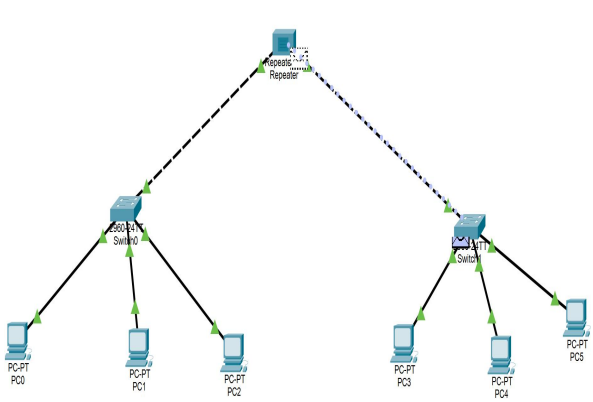
REPEATER

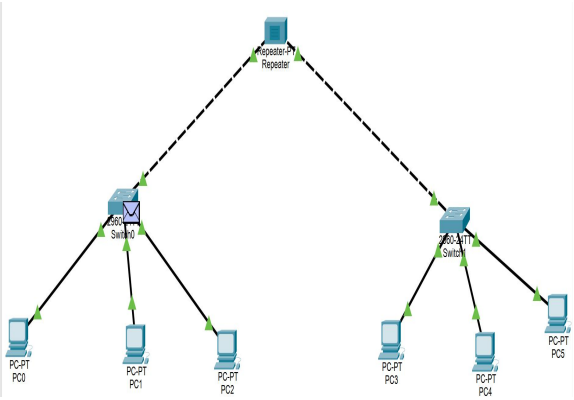












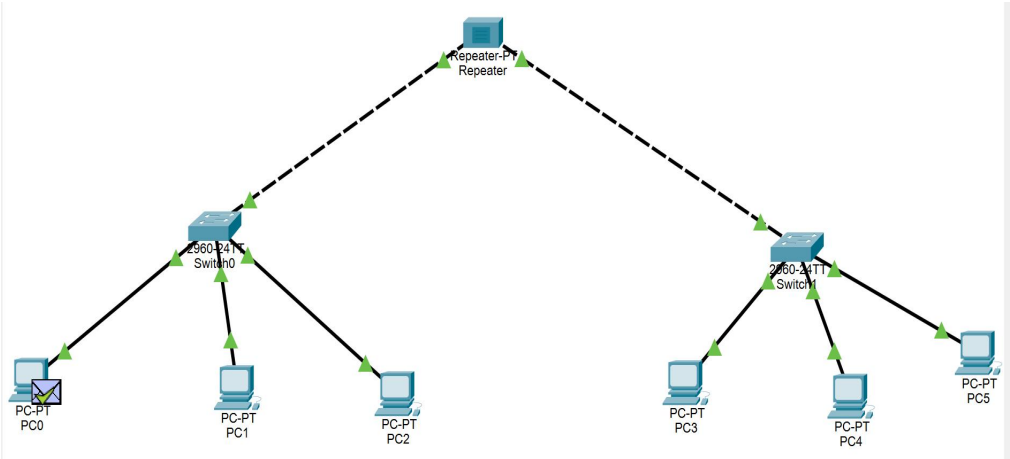


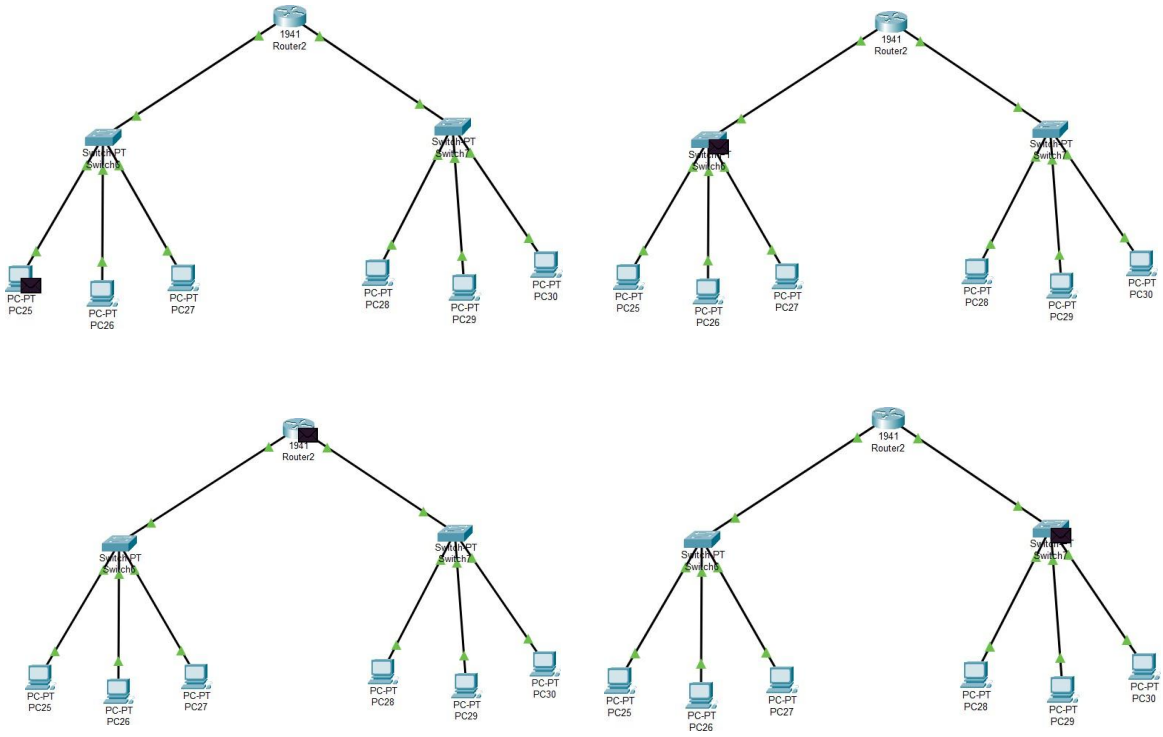
Fig: Repeater receiving a weak signal from the source, regenerating (amplifying) the signal, and forwarding it to extend the network distance without filtering data.

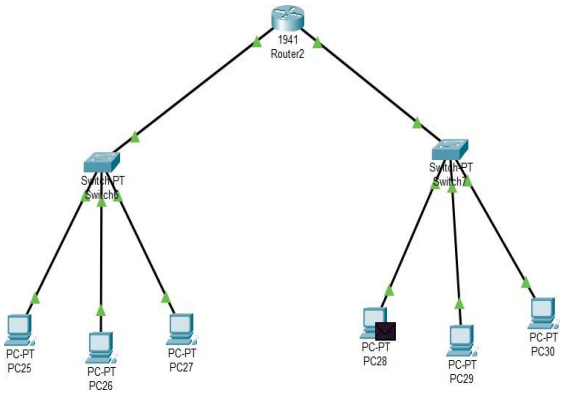
Configuration Table

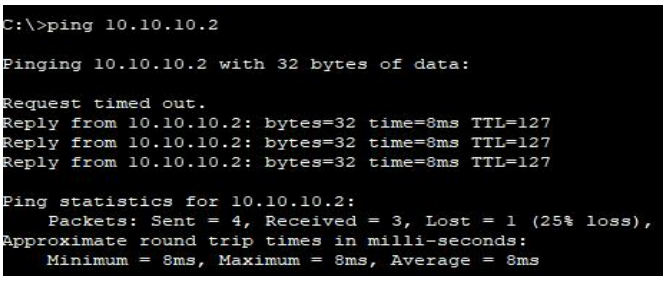
Device Name	Connection Type	Status
Repeater0	Port 0 to Left Segment	Active
Repeater0	Port 1 to Right Segment	Active

A Repeater was placed between two long-distance segments. The left network wire is connected to the Repeater Port 0, likewise the right network wire is connected to Repeater Port 1. The signal propagation was observed, ensuring that the packet could travel the extended distance without signal degradation

ROUTER







```
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Request timed out.
Reply from 10.10.10.2: bytes=32 time=8ms TTL=127
Reply from 10.10.10.2: bytes=32 time=8ms TTL=127
Reply from 10.10.10.2: bytes=32 time=8ms TTL=127

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 8ms, Average = 8ms
```

Switch: 2960-24TT Switch4 and 2960-24TT Switch5

Router: 1941 Router2	IP Address
GigabitEthernet0/0	192.168.1.1
GigabitEthernet0/1	10.10.10.1

PC Name	IP Address
PC-PT PC25	192.168.1.2
PC-PT PC26	192.168.1.3
PC-PT PC27	192.168.1.4
PC-PT PC28	10.10.10.2
PC-PT PC29	10.10.10.3
PC-PT PC30	10.10.10.4

Discussion And Conclusion

The networks were constructed by dragging end devices such as PCs and laptops, along with network devices including hubs, switches, and routers, onto the workspace in Cisco Packet Tracer. An important part of the setup was selecting the correct type of cabling. Copper straight-through cables were used to connect different types of devices, such as a PC to a switch, while copper cross-over cables were used to connect similar devices, such as switch-to-switch or PC-to-PC connections. Initially, the router simulation did not function correctly because the PCs in the second network were not configured with the correct default gateway. This issue was resolved by ensuring that the default gateway address of each PC matched the IP address of the corresponding router interface.

This lab provided a practical understanding of how different network devices handle data traffic. Through hub simulation, it was observed that a hub broadcasts data to all connected devices, leading to increased network traffic and security concerns. In contrast, the switch demonstrated intelligent data forwarding by using MAC address mapping, sending data only to the intended destination. The router played a vital role in enabling communication between different subnets by using IP addresses and default gateways. Overall, using Cisco Packet Tracer made it possible to clearly visualize and verify the functional differences between Layer 1, Layer 2, and Layer 3 devices without the risk of misconfiguring real hardware. The objectives of the lab were successfully achieved.